

RxGuard System Architecture & Specification

Prescription Dispensing & Real-Time Clinical Verification Platform

Statutory Standard: Ethiopian Standard ES 7084:2024
Authority: EFDA Good Dispensing Practice
Stack: Full-Stack MERN (Dockerized)
Date: September 2026 • Version 2.0

1. System Overview: How RxGuard Works

RxGuard is a mission-critical pharmaceutical operating platform developed in accordance with Ethiopian Standard **ES 7084:2024** and the Ethiopian Food and Drug Authority (EFDA) guidelines. The platform transforms manual retail pharmacy counters into high-reliability, automated clinical workstations by combining real-time artificial intelligence safety checks (<1.5s latency), automated First-Expiry-First-Out (FEFO) stock reservations, optical barcode verification (<300ms), and statutory bilingual (Amharic ፎጳ / English ግፆ) auxiliary thermal label generation.

End-to-End Prescription Lifecycle Workflow:

- 1 Prescription Intake & Digitization (UC01 / FR01-FR03)**
The counter operator selects or registers the patient profile and enters prescription line items. The system validates prescriber license regex (e.g. ETH-MD-XXXXX) and checks statutory order validity: **Acute orders ≤ 15 days** and **Chronic orders ≤ 30 days**. For pediatric patients under 12 years old, body weight is mandatory (DEF-05).
- 2 Real-Time Clinical Verification Engine (UC02 / FR04-FR06, PQR1 < 1.5s)**
Upon submission, the deterministic AI safety engine cross-analyzes the regimen across four clinical safety vectors: **(a) Drug-Drug Interactions (DDI)** (e.g. Warfarin + Aspirin critical bleed risk, Ciprofloxacin + Theophylline clearance inhibition), **(b) Documented Allergies** (e.g. Amoxicillin in penicillin-allergic patients), **(c) Dose Boundary Limits** (mg/kg/day\$ pediatric formula checks), and **(d) Disease Contraindications** (e.g. NSAIDs in active peptic ulcer disease). If critical flags exist, the prescription is placed on *Clinical Hold*.
- 3 Clinical Override Authorization (UC03 / DEF-02)**
Critical and moderate safety alerts can only be overridden by an authorized **Licensed Pharmacist**. The pharmacist must enter a statutory clinical justification of **at least 15 characters** and sign with a dedicated 4-digit PIN (1234). All overrides are immutably archived into the audit trail with timestamps and operator credentials.
- 4 Physical Inventory Staging & FEFO Pick-List (UC04 / FR07-FR08)**
Once clinically cleared, the system executes First-Expiry-First-Out (FEFO) allocation, guiding the technician to the exact shelf location of the lot expiring earliest. The **30-Day Expiry Guard (FR08 / DEF-01)** automatically halts reservation of any batch expiring in ≤ 30 days unless supervisory clearance is recorded.
- 5 Optical Barcode Scan Matching (UC05 / FR09, PQR2 < 300ms)**
At the dispensing counter, the operator scans the physical 1D UPC or 2D GS1 DataMatrix barcode on the drug package. The system matches molecule, strength, and batch in <300ms. If a mismatch is detected (e.g., scanning Metformin 850mg instead of prescribed 500mg), a loud audio alarm sounds and a full-screen blocking modal freezes label printing until rectified.
- 6 Final Handover & Bilingual Label Generation (UC05 / FR10, FR12)**
Once 100% of line items are optically verified, the Finalize button unlocks. The engine decrements perpetual stock on hand in real time, generates Ethiopian Standard ES 7084:2024 compliant bilingual thermal labels (Amharic ፎጳ and English ግፆ) containing dosage and storage instructions, and outputs a customer POS thermal receipt.

2. Professional Roles Specification & Strict RBAC

RxGuard enforces statutory Role-Based Access Control (RBAC) required by Ethiopian regulatory frameworks. Operators must select their certified role at the **Login Gateway** prior to authentication. In-session role switching is prohibited to guarantee clinical non-repudiation and prevent unauthorized scope escalation.

 Licensed Pharmacist CLINICAL AUTHORITY Demo User: Sitra Temam, BPharm (sitra_pharmacist) Credential: License ETH-PH-70841 • PIN: 1234 The primary clinical practitioner holding legal responsibility for patient safety and drug therapy review under ES 7084:2024. Authorized Privileges: - Review and approve clinically flagged prescriptions. - Execute clinical overrides with 15+ char rationale & PIN. - Log prescriber phone clarifications. - Authorize near-expiry batch emergency release (FR08). - Dispense medications and issue bilingual labels. Restricted (Cannot Do): ✗ Cannot register new patients or add new drugs to the master formulary (reserved for Administrator to prevent unverified additions).	 Pharmacy Technician DISPENSING & STOCK Demo User: Abebe Kebede (abebe_tech) Credential: License ETH-PT-40281 The technical operations specialist managing counter intake entry, physical inventory picking, and optical scanning. Authorized Privileges: - Digitize written prescription slips into the system. - View FEFO stock allocation pick-lists and shelf locations. - Perform optical barcode scanning on drug packages. - Finalize dispensing for clinically approved orders. - Search drug monographs in read-only mode. Restricted (Cannot Do): ✗ Cannot override DDI or safety alerts (no PIN authority). ✗ Cannot register patients or add drugs to catalog. ✗ Cannot trigger external EFDA formulary synchronization.	 System Administrator MASTER DATA GOVERNANCE Demo User: Dawit Haile (dawit_admin) Credential: Credential ETH-SYS-ADMIN The governance authority responsible for pharmacy master data, patient registry, regulatory formulary, and audit oversight. Authorized Privileges: Register New Patients: Create demographic profiles with allergies, chronic illnesses, and pediatric weights. Add New Medications: Add drugs with strengths, bounds, unit pricing, auto-provisioned FEFO batches & barcodes. - Trigger EFDA portal external monograph synchronization. - Inspect immutable regulatory audit trails & narcotics registers. Restricted (Cannot Do): ✗ Does not substitute for the licensed pharmacist in signing off clinical overrides during counter dispensing.
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3. RBAC Matrix: Feature Permissions by Role

SYSTEM FEATURE / FUNCTIONALITY	LICENSED PHARMACIST	PHARMACY TECHNICIAN	SYSTEM ADMINISTRATOR
Prescription Order Intake (UC01)	✓ Full Access	✓ Full Access	✓ Full Access
Clinical AI Verification Review (UC02)	✓ Full Clinical Authority	✗ View Only (Cannot Clear)	✓ Full Audit Review
DDI & Safety Override with PIN (UC03)	✓ Authorized (PIN 1234)	✗ Locked (No Authority)	✗ Clinician PIN Required
FEFO Stock Staging & Pick-List (UC04)	✓ Authorized	✓ Primary Handler	✓ Audit & Stock View
Near-Expiry Batch Release Override (FR08)	✓ Supervisory Override	✗ Blocked per DEF-01	✓ Supervisory Override
Optical Barcode Verification (UC05)	✓ Authorized	✓ Primary Operator	✓ Test & Audit Mode
Bilingual Label & Receipt Print (FR10)	✓ Authorized	✓ Primary Operator	✓ Reprint / Audit
Register New Patients (Master Data)	✗ Hidden (Admin Only)	✗ Hidden (Admin Only)	✓ Exclusive Authority
Add New Drugs to Formulary (Catalog)	✗ Hidden (Admin Only)	✗ Hidden (Admin Only)	✓ Exclusive Authority
Trigger EFDA Regulatory Sync (UC06)	✓ Permitted	✗ Restricted (US-PA-02)	✓ Primary Authority
Regulatory Audit Logs & Narcotics (FR13)	✓ Clinical Register View	✗ Restricted	✓ Full Immutable Ledger

4. Why In-Session Role Switching is Prohibited

Regulatory Mandate (ES 7084:2024 / EFDA Article 14): In clinical and dispensing environments, an operator must authenticate under their own distinct credentials. In-session role dropdowns (such as switching from Technician to Pharmacist with a simple click) violate non-repudiation standards because they allow unauthorized staff to perform clinical overrides or master data modifications under another operator's identity without re-authentication.

In RxGuard: **(1) Role selection is made at the login gateway** before credentials are submitted; **(2) Once authenticated, the role is cryptographically bound to the JWT session token** and permanently locked inside the interface; **(3) To change roles, the operator must explicitly click "Sign Out"**, clearing the active session token and requiring the new operator to provide valid credentials and PIN for their specific regulatory persona.